



**Politecnico
di Torino**

Film-Driven Brain Topology

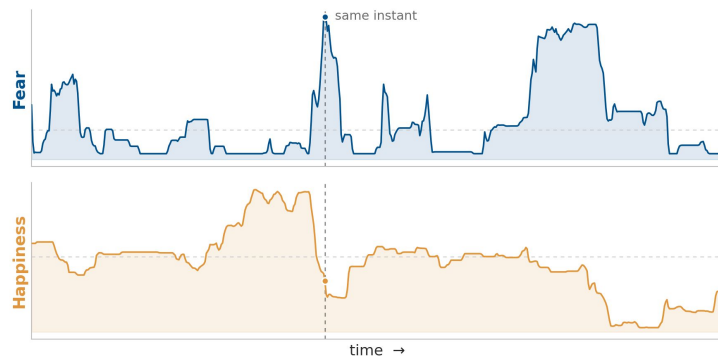
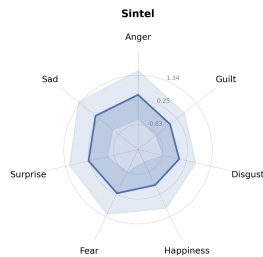
A Multi-Scale Mapper Analysis of fMRI Emotional Dynamics

Candidate:
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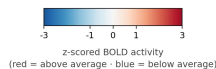
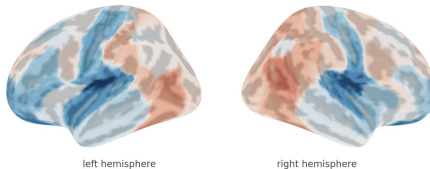
Supervisors:
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How does the brain change when we experience emotions while watching a film?

“Sintel”, 2010



Sintel — cortical activity at t = 235 (fear peak)



Is there a link between emotions and brain activity?

The data: Emo-Film



30 subjects
(2 excluded)



14 films
1 rest session



44 annotators

fMRI sessions

- 420 (15*28)
- Schaefer atlas
- 100 ROI

X: time (TR) x regions of interest (ROI)

X	ROI 1	...	ROI 400
TR1			
...			

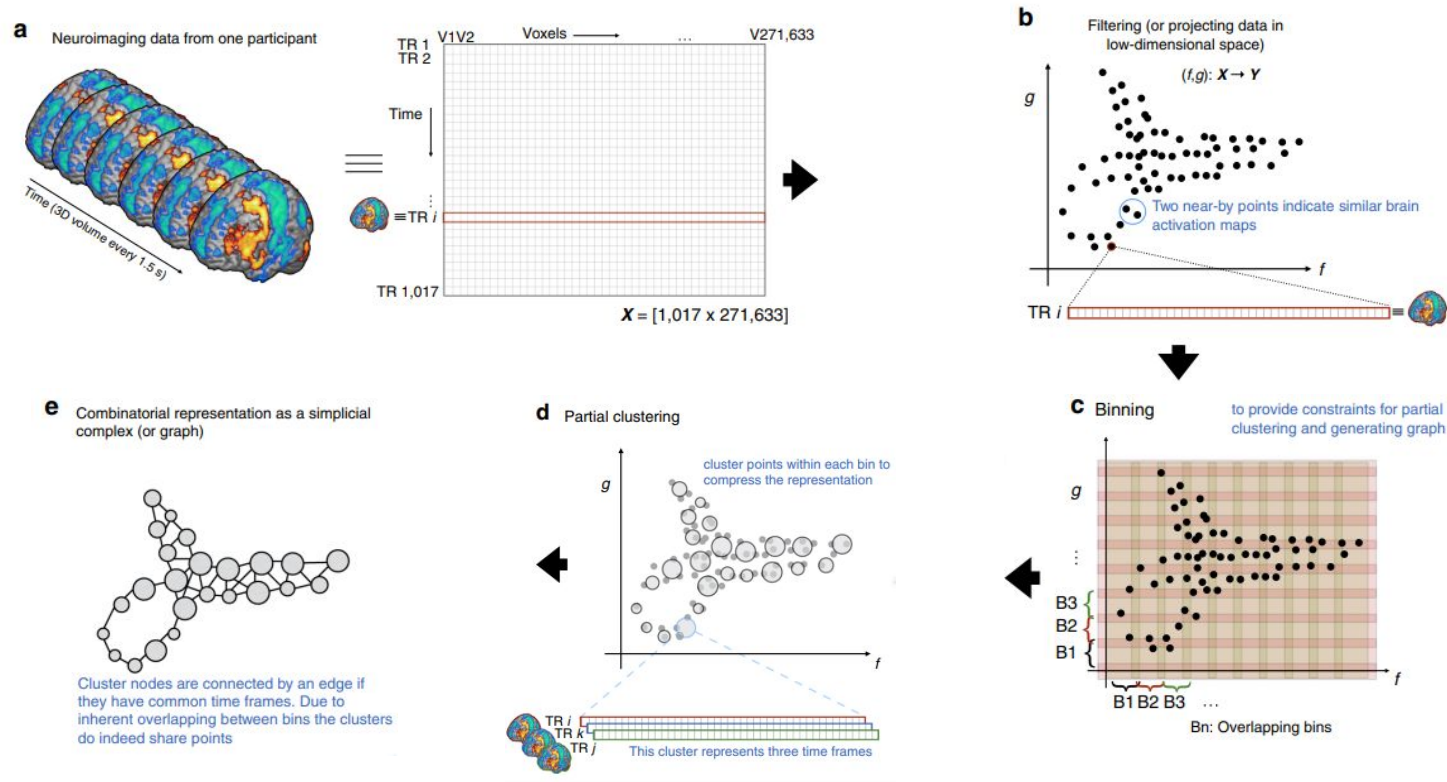
Emotion annotation

- 50 features
- 44 annotations averaged

E: time (TR) x emotional features

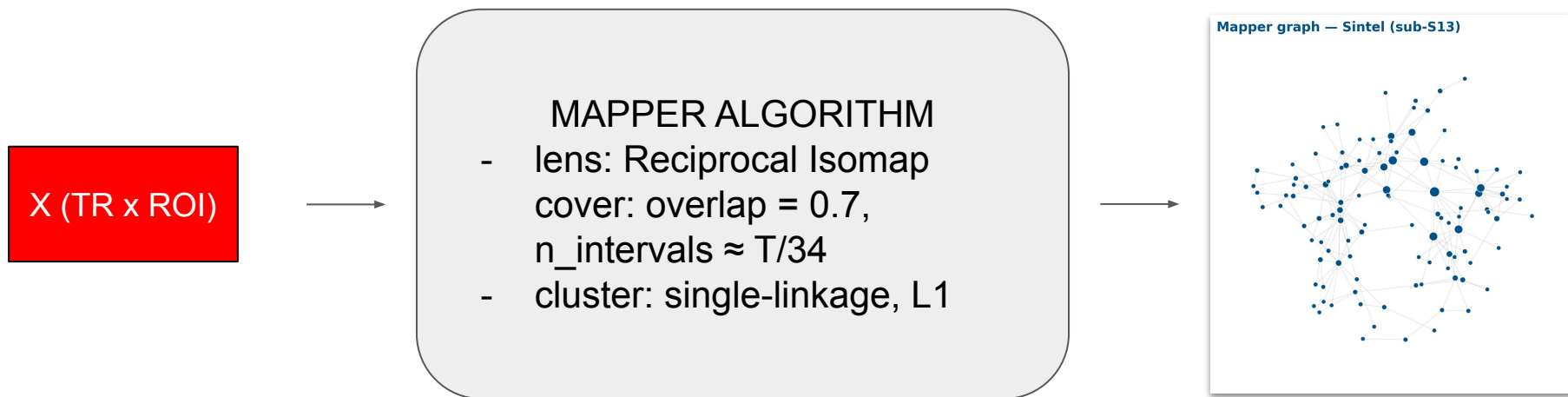
E	feature 1	...	feature 50
TR1			
...			

The tool: the Mapper algorithm



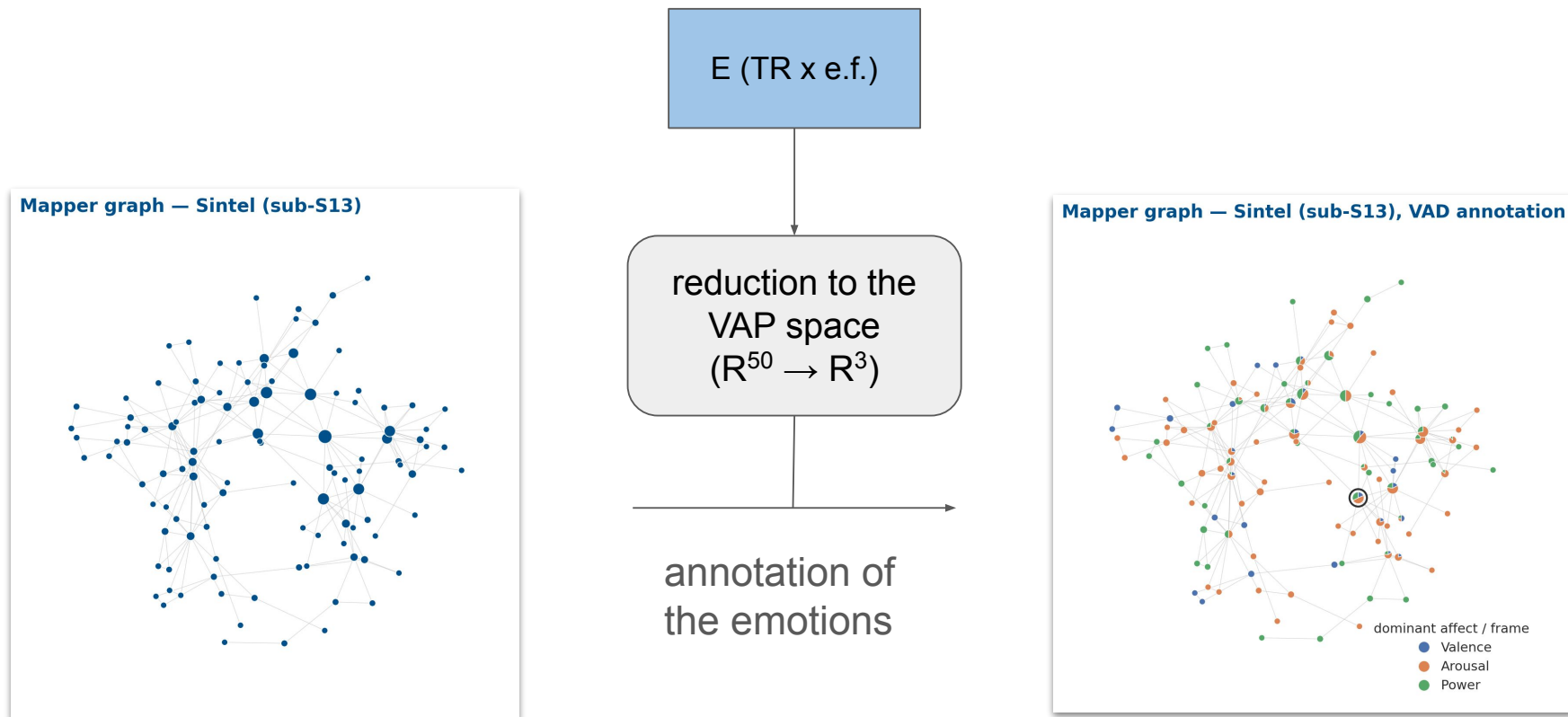
The Pipeline: from High-Dimensional Data to Graphs

for each (subject-film) couple:



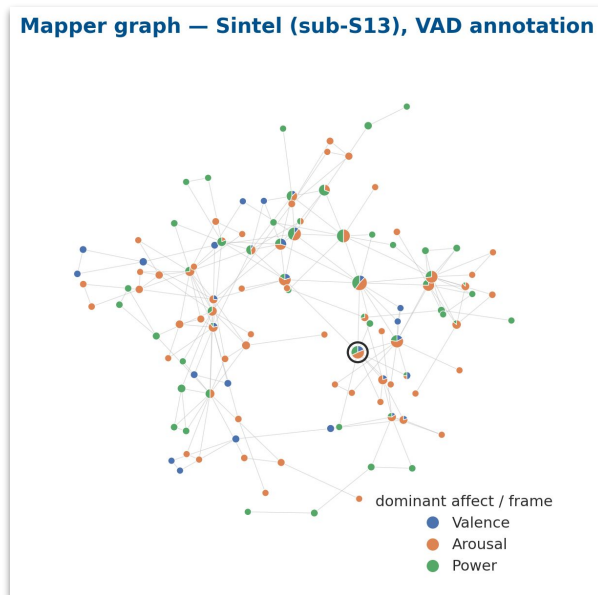
r. isomap: Van der Maaten & Hinton (2008), Journal of Machine Learning Research
t-SNE: Liao & Carin (2009), ICML
parameters: Saggar et al. (2022), Nature Communications

The Pipeline: from High-Dimensional Data to Graphs



Analysis Overview

- 420 graphs (one for each couple)
- 28 avg graphs, 28 rest graphs



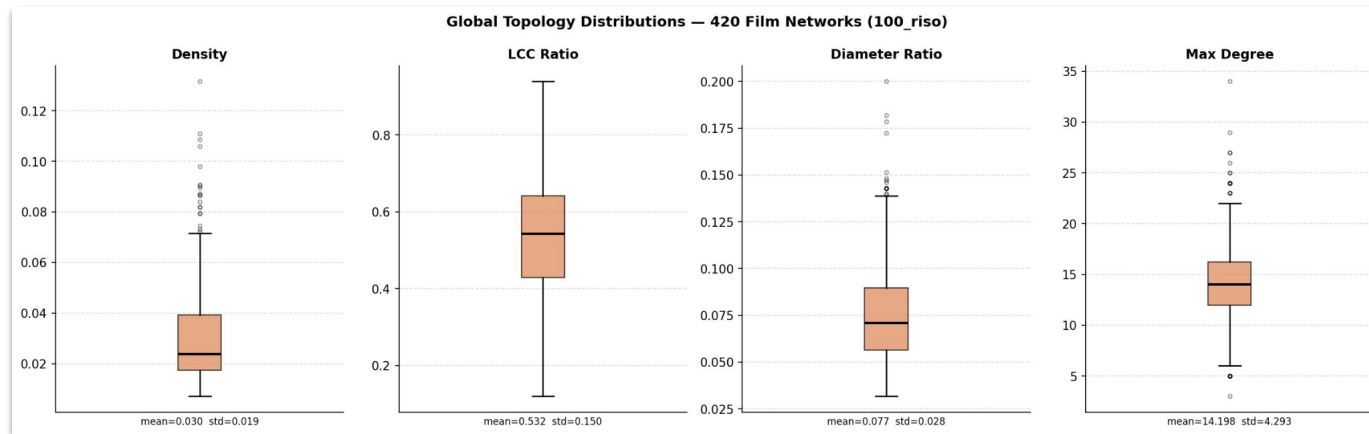
(I) Macro Scale

(II) Meso Scale

(III) Micro Scale

(I) Macro Scale

- Density
- LCC Ratio
- Diameter Ratio
- Max Degree

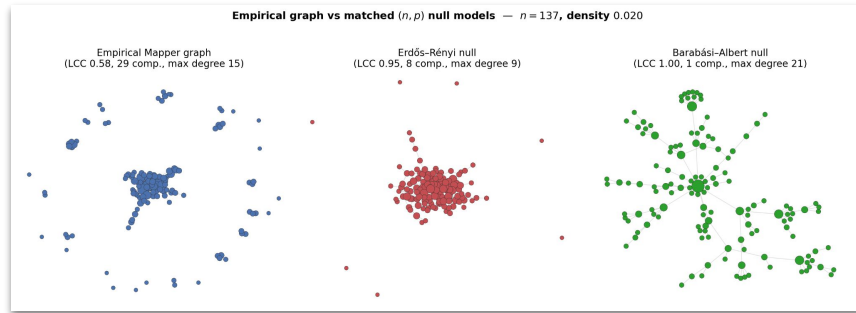


Rest vs Film:
Wilcoxon → n.s.

Density	LCC Ratio	Diameter Ratio	Max Degree
p = 0.6	p = 0.058	p = 0.080	p = 0.38

(I) Macro Scale

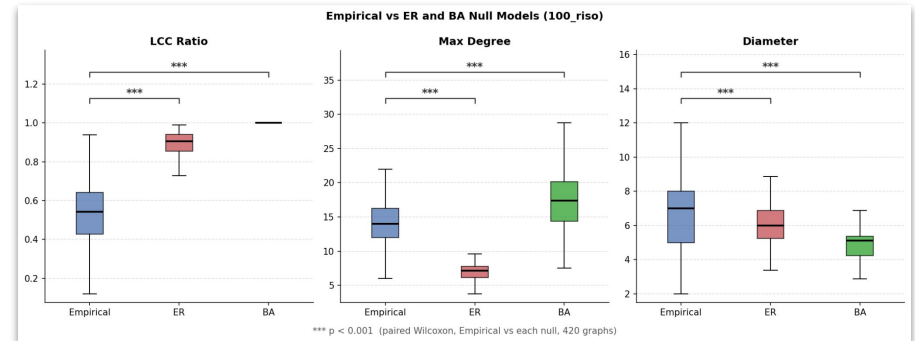
Null Models (ER, BA)



Erdős & Rényi (1959), Publicationes Mathematicae
Debrecen

Barabási & Albert (1999), Science

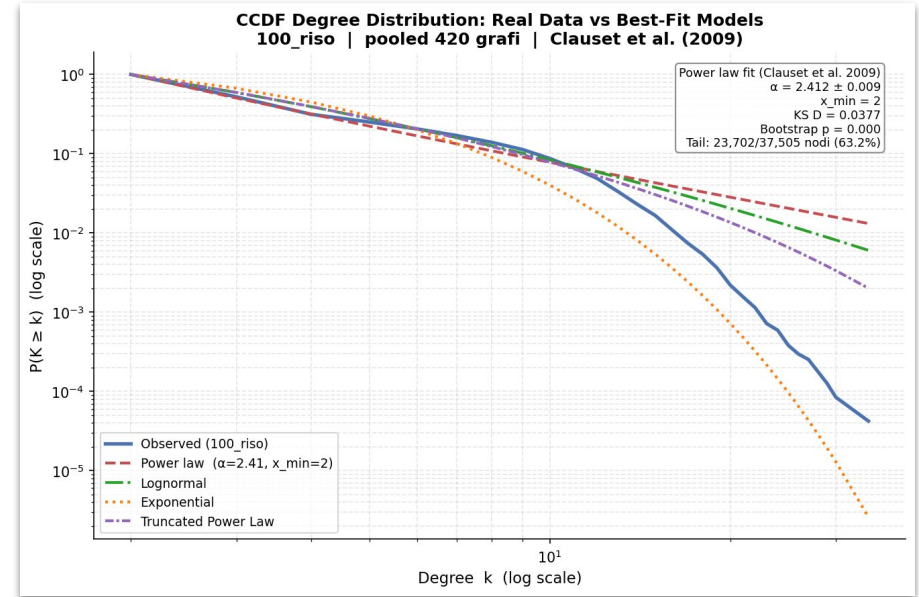
Null Models vs Film: Wilcoxon



(I) Macro Scale

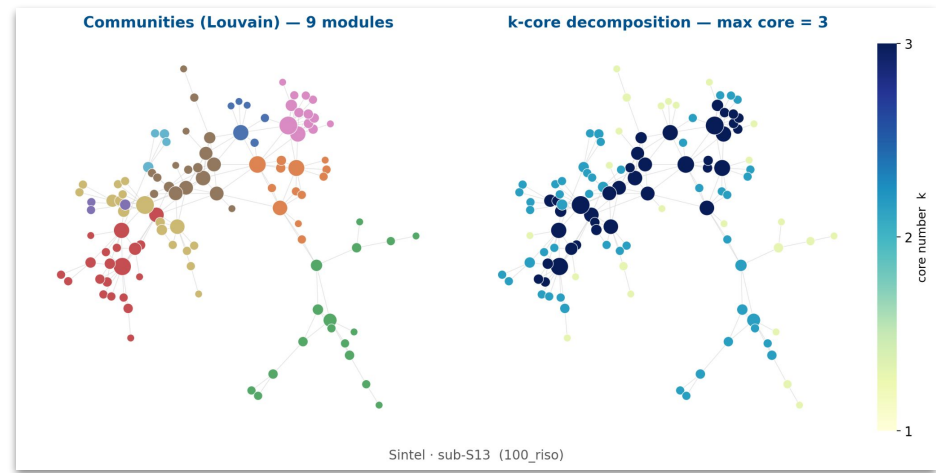
Degree distribution – Power Law Test

- 37 505 pooled nodes
- pure power-law rejected
- best fit truncated power-law



(II) Meso Scale

- k-core decomposition
- communities detection (Louvain algorithm)



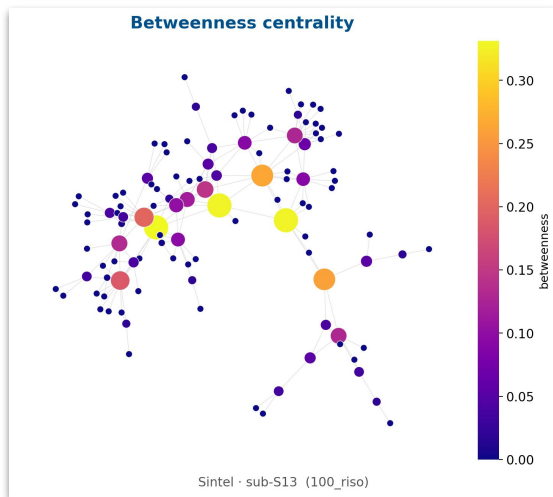
Louvain: Blondel et al. (2008), Journal of Statistical Mechanics

Rest vs Film
Wilcoxon

N Communities	Max k-core	Frac. innermost
film > rest	film > rest	rest > film
p = 0.015 *	p = 0.020 *	p = 0.017 *

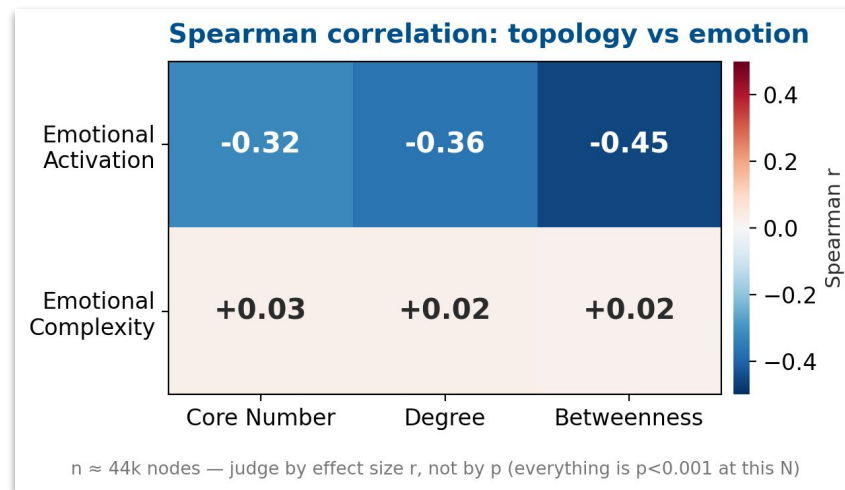
(III) Micro Scale

- betweenness centrality



- emotional activation → intensity
- emotional complexity → diversity

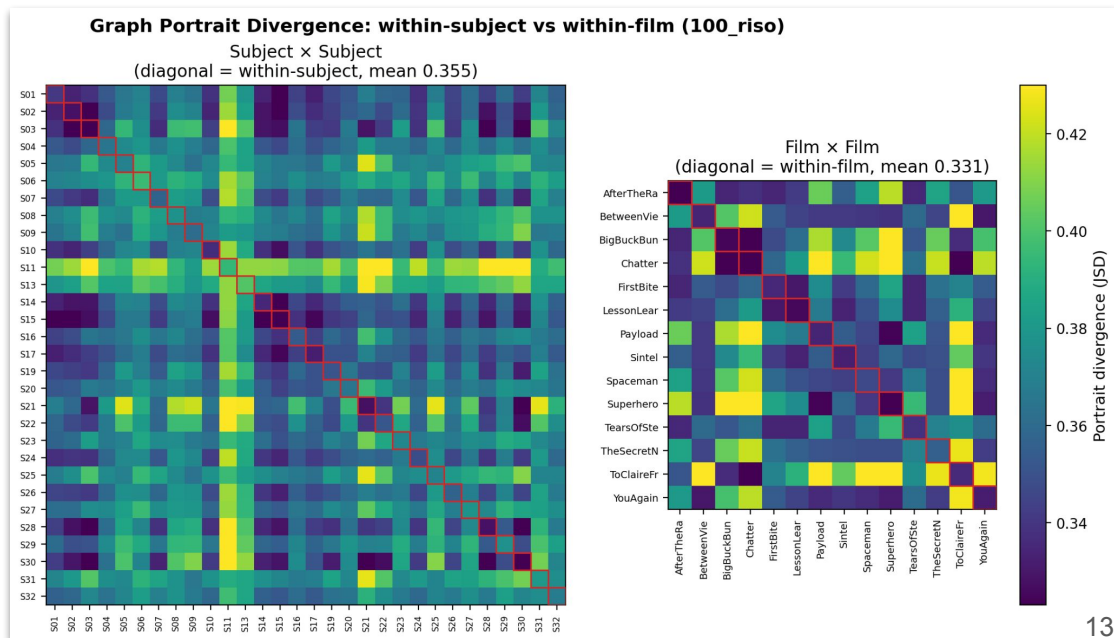
44000 nodes pooled



- the more central a brain state, the fewer emotions are strongly active
- emotional diversity does not depend on the network properties

(IV) Whole-graph: film entertainment vs individual signature

- Graph portrait: matrix $B[l, k]$ = number of nodes with k nodes at distance l
- Portrait divergence: Jensen-Shannon distance between portraits
- within film JSD 0.331 < within-subject 0.355
- subject identification rate 4.9% vs chance 3.3%



Conclusions

- Scale matters
- Emotional Activation
depends on the shape of
the graph

Future perspectives

- Extend the whole analysis to all
of the configurations
- Graph Neural Networks

Thank you for listening!

Code available at: <https://github.com/davide-raciti/tda-brain-thesis/tree/main>